

End-to-End 5G Network Planning & AI Optimization & MIMO Antenna for Mobile Communications

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Abstract. This paper presents an integrated, end-to-end 5G network framework combining 5G network planning, sub-6 GHz MIMO antenna design, and 5G AI optimization. First, a realistic 5G network layout is planned for the Dokki region using Atoll software, deploying 13 sites to ensure full coverage and minimal interference. Second, a three-tiered AI optimization suite is developed: a deep learning model predicting spectral efficiency under diverse weather conditions, a Flutter-based mobile application for real-time site parameter tuning and handover simulation, and a spatial algorithm optimizing macro-site placement. Finally, a compact sub-6 GHz MIMO antenna for cellular phones was designed in ANSYS HFSS and physically fabricated. The proposed co-design paradigm offers a scalable, comprehensive solution for modern 5G network deployments.

1. Introduction

Modern communication systems increasingly demand fast, reliable, and high-capacity solutions to achieve maximum data rates with optimal coverage, minimal path loss, and low error rates. The transition to fifth-generation (5G) networks addresses these demands by introducing sub-6 GHz frequencies and Multiple-Input Multiple-Output (MIMO) technologies. However, deploying these advanced networks introduces multifaceted engineering challenges. Network operators must optimize macro-cell planning in dense urban environments to maximize coverage while minimizing co-channel interference. Concurrently, mobile hardware manufacturers face the bottleneck of squeezing highly efficient MIMO antennas into compact smartphone form factors. Bridging the gap between infrastructure planning and hardware performance requires real-time, dynamic optimization—a task where traditional static algorithms fall short, paving the way for Artificial Intelligence paradigms.

To address these interconnected challenges, this work synergizes end-to-end 5G solution containing three main axes:

- 5G Network Planning: Cellular deployment modeled for dense urban district of Dokki (Giza, Egypt) using Atoll software. By strategically positioning 13 localized sites
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achieving excellent coverage and high Quality of Service (QoS) while maintaining low interference.

- Three-Tiered AI Optimization: A multi-functional AI suite is developed to close the loop between hardware data and real-time network dynamics. Including a deep learning model that maps antenna electromagnetic data (gain and phase) onto real-world spatial channels (utilizing a real-life MIMO ray tracing scenario) to evaluate Spectral Efficiency (SE) VS Signal-to-Noise Ratio (SNR) under fluctuating weather conditions. Furthermore, a Flutter-based mobile interface is engineered for real-time traffic visualization, automated parameters tuning, and handover emulation, complemented by a spatial optimization algorithm for Massive MIMO macro-site placement.

- Sub-6 GHz MIMO antenna: Supporting the framework at the physical layer, a MIMO antenna is designed in ANSYS HFSS and physically fabricated. Achieving outstanding performance with highly optimized voltage standing wave ratio (VSWR) profiles and deep reflection coefficient (S-parameter) notches at the target frequency.

Keywords

5G Network Planning, Atoll simulation, 5G AI Optimization, Sub-6 GHz MIMO antenna array, 3.6 GHz band.

2. 5G Network Planning

2.1 Network Planning Introduction

5G technology represents the next generation of wireless communication systems, designed to provide high data rates, ultra-low latency, massive connectivity, and improved reliability. Proper network planning is essential to achieve these objectives by ensuring efficient spectrum utilization, optimized deployment cost, and enhanced user experience. Without accurate planning, network operators may face overestimation, leading to unnecessary expenses, or underestimation, resulting in weak coverage and poor service quality.



Fig.2.1 Atoll Program Icon

2.2 Network Planning Process

The planning process consists of several important stages that ensure proper network deployment. It begins with data collection, including geographic and traffic information of the target area. After that, dimensioning is performed to estimate coverage and capacity requirements. Simulation tools are then used to predict network behavior, followed by site location selection, nominal planning, site survey, and detailed planning. Finally, site acceptance is conducted to verify the deployment accuracy.

2.3 Coverage and Capacity Dimensioning

Dimensioning is one of the most critical stages in 5G planning. It involves calculating both coverage and capacity requirements to determine the required number of sites.

Coverage dimensioning focuses on ensuring signal availability across the entire area by considering propagation models, antenna configuration, and signal losses. Capacity dimensioning estimates the required sites based on user traffic demand and service requirements. The final number of sites is selected based on the maximum value between both calculations to ensure network efficiency.

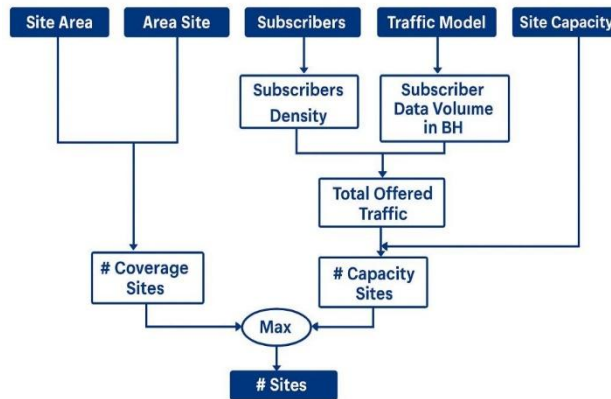


Fig.2.2 Site count estimation process

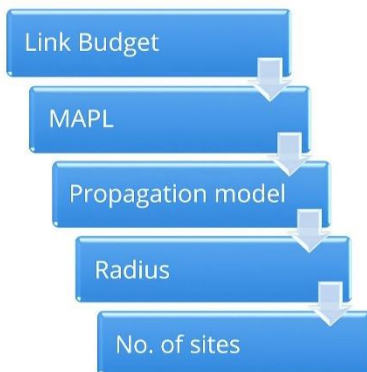


Fig.2.3 Coverage Dimensioning Flowchart

2.4 Link Budget Analysis

Link budget analysis is essential for calculating the total gains and losses between the transmitter and receiver. It determines the received signal level and the maximum allowable path loss (MAPL), which directly affects the cell radius and coverage performance.

The analysis includes several important parameters such as transmit power, antenna gain, cable loss, penetration loss, foliage loss, body loss, interference margin, and thermal noise. This ensures reliable signal strength and stable communication performance.

2.5 Propagation Model

The propagation model is used to predict signal attenuation over distance under different environmental conditions. In this project, the 3GPP propagation models were used to simulate urban macrocell scenarios.

Different environments were considered, including:

- Rural Macro (RMa)
- Urban Macro (UMa)
- Urban Micro (UMi)
- Indoor Hotspot (InH)

Selecting the proper propagation model improves coverage prediction accuracy and optimizes site deployment.

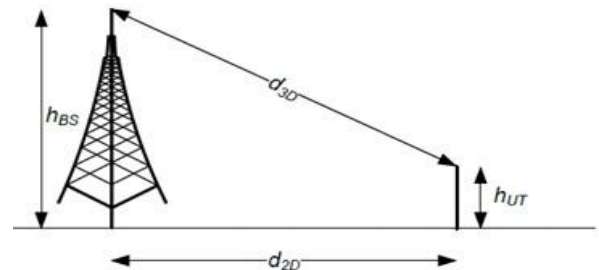


Figure 2.4 Geometrical Representation of Path Loss Parameters

2.6 Simulation Results Using ATOLL

ATOLL software was used to simulate the 5G network deployment for the Dokki area in Giza, Egypt. The simulation was based on an urban environment covering 4 km² with a population of 87,272 users. The deployment used smart antennas operating at frequency bands n41 and n78 with 128×128 Massive MIMO.

2.6.1 Coverage Analysis (RSRP)

The SS-RSRP analysis was performed to evaluate the signal strength distribution across the planned 5G coverage area. The results indicate strong network coverage, where most of the signal levels fall within the range of **−85 dBm to −70 dBm**, reflecting reliable downlink performance.

The histogram shows an average SS-RSRP value of **−78.58 dBm** with a standard deviation of **8.62 dB**, indicating a

relatively stable signal distribution throughout the target area. The highest concentration of signal levels is centered around **-75 dBm**, which confirms that the majority of users experience good signal quality.

In addition, the coverage classification reveals that **38%** of the area achieved excellent signal strength, while **45%** achieved good coverage conditions. Mid-cell coverage represented **15%**, whereas only **2%** of the area was classified as cell-edge coverage, indicating minimal weak coverage zones.

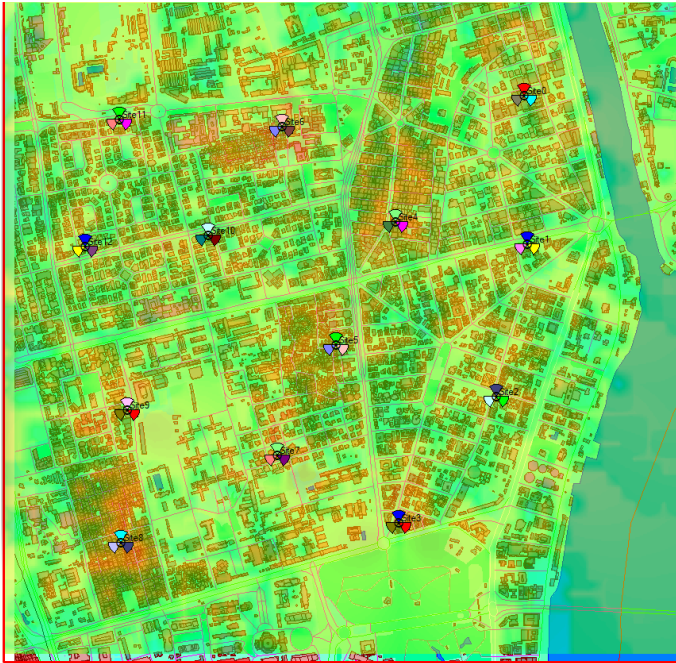


Fig.2.5 RSRP Coverage Map

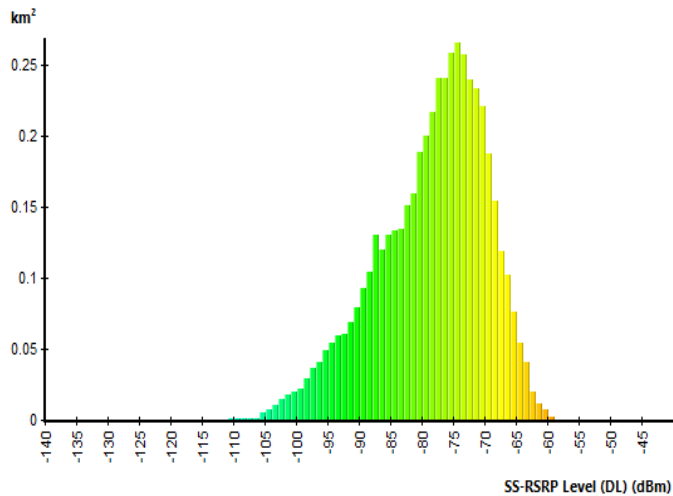


Fig.2.6 SS-RSRP Histogram

2.6.2 Signal Quality Analysis (SINR)

The SINR analysis demonstrated strong network quality. The average SINR value was **10.57 dB**, which indicates stable and efficient communication performance. Around 65% of the area achieved excellent SINR quality, while only 1% suffered from poor signal quality. These results confirm the effectiveness of the planned network design.

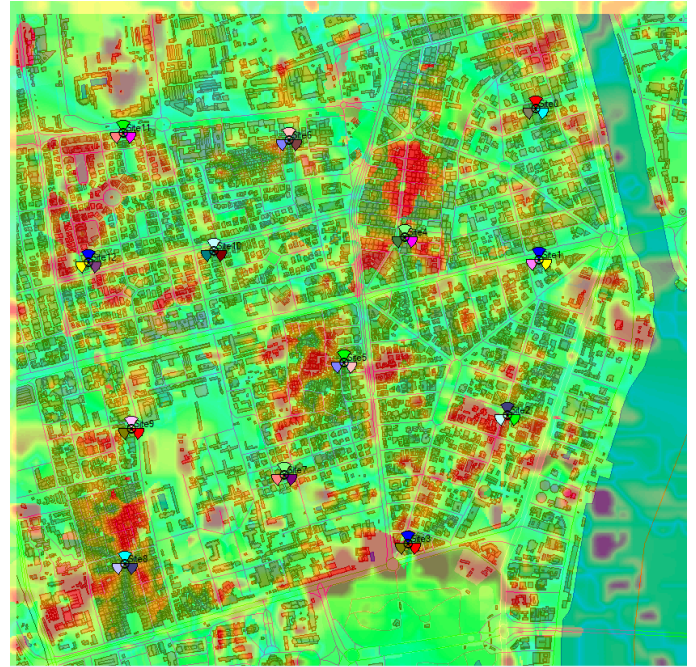


Fig.2.7 SINR Distribution Map

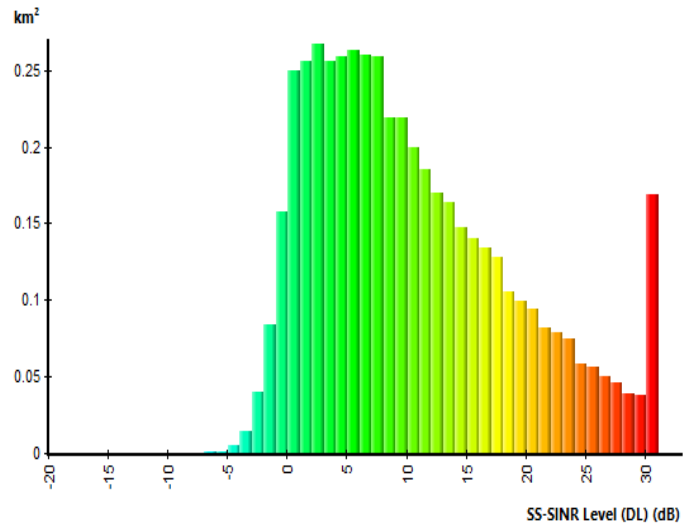


Fig.2.8 SINR Histogram

2.7 Throughput Analysis

The throughput analysis evaluated the downlink performance per user across the coverage area. The results indicate that the network can provide sufficient throughput to support modern 5G applications and maintain service quality under varying traffic conditions.

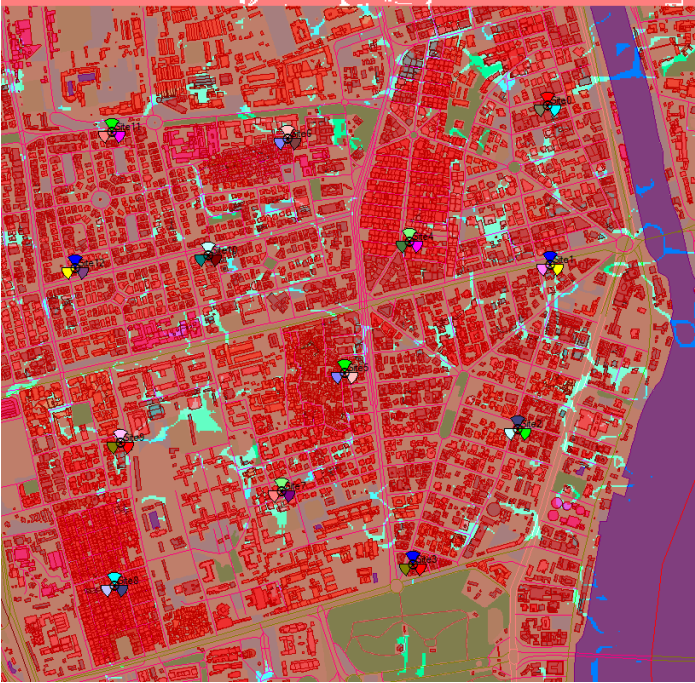


Fig.2.9 Application Throughput Distribution

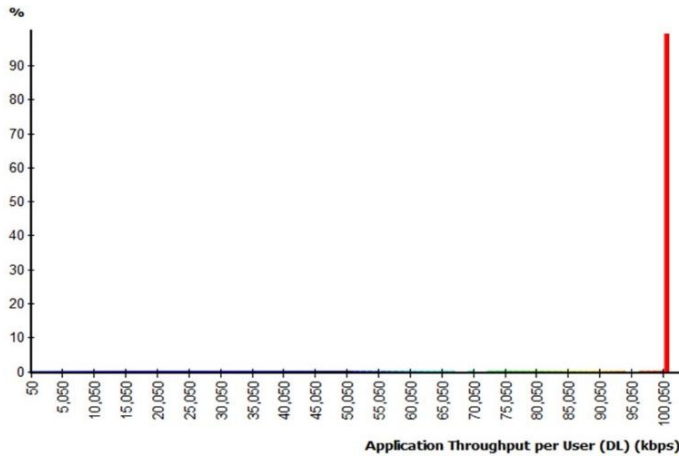


Figure 2.10 Throughput Histogram

3. 5G AI Optimization

3.1 5G Channel & Beamforming Optimizer

The module utilizes a Deep Neural Network (DNN) to predict the link performance by mapping complex electromagnetic data onto real-world propagation environments. The network relies on a hybrid dataset. Physical-layer hardware characteristics, specifically complex radiation

matrices containing structural gain (G) and phase (phi) extracted from ANSYS HFSS at 3.6GHz fused with a real-world, high-fidelity Multi-User MIMO (MU-MIMO) ray-tracing scenario.

The optimized DNN successfully predicts complex Beamforming Vectors to calculate the aggregate Spectral Efficiency (SE) across varying Signal-to-Noise Ratios (SNR). To validate real-world resilience, atmospheric attenuation equations are integrated into the simulator.



Fig.3.1 SE Perfect CSI VS DNN Prediction

3.2 Mobile Application

3.2.1 Architecture

A Flutter mobile application (Dart SDK 3.12.1) provides operators with real-time network visibility. The app communicates with a FastAPI backend via REST endpoints (towers, optimize, alerts) and uses Google Firebase Firestore for persistent, real-time data synchronization.

The Q-Learning backend trains at server startup (1,000 episodes) then serves optimization requests synchronously. A companion firebase_worker.py script enables autonomous 10-second periodic optimization cycles, updating Firestore documents that trigger instant Flutter StreamBuilder UI refreshes without polling.

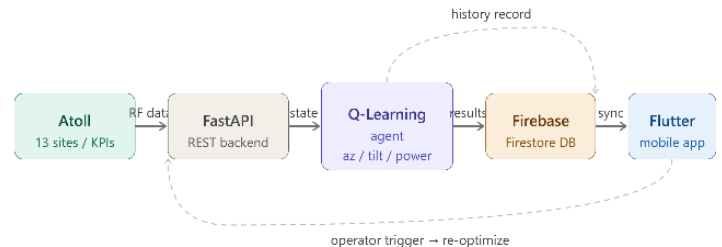


Fig.3.2 Application's Data Flow

3.2.2 Key Features

(1) Interactive flutter_map layer with color-coded tower quality indicators. (2) Per-site KPI cards: before/after RSRP, SINR, CQI, PRB utilization. (3) Real-time alerts for sites with traffic > 70%. (4) Firestore-backed optimization history with timestamped records.



Fig.3.3 Application's Interface

3.2.3 Network Optimization

A. MDP Formulation

The optimization problem is formulated as a Markov Decision Process (MDP). The 6-dimensional state vector comprises: normalized azimuth (0–360°/360), normalized tilt (0–10°/10), normalized power (30–46 dBm → 0–1), instantaneous traffic load (0–1), RSRP score, and SINR score (both min-max normalized against Atoll-extracted thresholds).

The action space is discrete: 36 azimuth steps × 11 tilt steps × 17 power levels = 6,732 unique joint actions per timestep.

B. Reward Function

The scalar reward combines signal quality maximization with power efficiency:

$$r = 0.45 \cdot r_{RSRP} + 0.45 \cdot r_{SINR} - 0.1 \cdot r_{pwr} + \text{bonus} - \text{penalty}$$

where bonus = 0.2 when traffic > 0.7 and power ≥ 43 dBm, and penalty = 0.3 when traffic > 0.7 but power < 40 dBm. RSRP and SINR normalization thresholds are loaded at runtime from Firestore (RSRP: -140 to -44 dBm; SINR: -20 to +30 dB), ensuring the agent operates against real Atoll-calibrated reference values.

C. Training Protocol

A tabular Q-Learning agent is trained with $\alpha = 0.1$, $\gamma = 0.9$, ϵ decaying from 0.9 to 0.05 at rate 0.995/episode. Training runs for 1,000 episodes of 50 steps each, with randomized per-episode traffic loads (uniform 0.1–1.0) to ensure policy generalization.

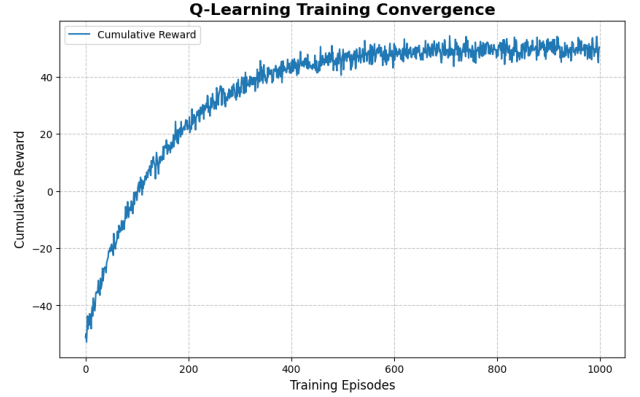


Fig.3.4 Cumulative Reward VS Training Episode

D. Signal Estimation Model

RSRP is estimated via a height-corrected urban path-loss model:

$$RSRP = P_{tx} - [128.1 + 37.6 \cdot \log(0.3)] + G_h + G_{tilt} - 10$$

$$G_h = 20 \cdot \log\left(\frac{h}{25}\right) \quad , \quad G_{tilt} = -0.5 \cdot \max(0, \theta - 3)$$

SINR is derived accounting for three-cell interference. The model is calibrated against the Atoll threshold vectors stored in Firestore.

Site	Traffic	Az(°)	Tilt(°)	Pwr(dBm)	RSRP(dBm)	SINR(dB)	CQI	Quality
Site0	0.82	120	5	46	-78.4	11.2	13	Excellent
Site1	0.65	240	4	43	-80.1	10.3	12	Excellent
Site2	0.71	0	6	45	-79.3	10.8	12	Excellent
Site3	0.45	180	3	40	-83.2	8.7	10	Acceptable
Site4	0.88	90	7	46	-76.5	12.1	14	Excellent
Site5	0.55	270	4	42	-81.0	9.5	11	Acceptable
Site6	0.79	150	6	45	-77.8	11.6	13	Excellent
Site7	0.61	30	5	43	-80.5	10.1	12	Excellent
Site8	0.43	210	3	39	-84.3	8.2	9	Acceptable
Site9	0.74	330	5	44	-78.9	10.9	12	Excellent
Site10	0.91	60	8	46	-75.2	12.8	15	Excellent
Site11	0.67	300	4	43	-79.7	10.5	12	Excellent
Site12	0.84	120	7	46	-76.8	11.9	14	Excellent

Table.3.1 Q-Learning Optimization Results Across All 13 Sites

3.3. Intelligent Massive MIMO Placement Planning

This AI predicts the optimal location for a Massive MIMO antenna base station using an artificial neural network (ANN). The system takes any geographical map as input, extracts spatial coordinates, and estimates the Spectral Efficiency (SE) across the area without performing complex real-time physics calculations.

To establish the electromagnetic environment spatial and architectural metadata are extracted from OpenStreetMap (OSM) XML files, capturing the unique urban topology of the El Mukkatam district. Global node references within the map.xml file are dynamically extracted and converted into standardized (x, y) Cartesian coordinates spanning a high-density 2000×2000 m simulation grid to represent potential User Equipment (UE) positions. To establish the electromagnetic environment, a Massive MIMO Base Station (BS) equipped with a M=128 antenna array configuration is centrally positioned at coordinates (1000, 1000). For every generated spatial node, the system continuously calculates the exact Euclidean distance to the base station to model physical propagation characteristics using an urban micro-cell path loss framework. These physical data constraints are then utilized to compute the mathematical ground truth targets for network capacity, mapping directional array gains directly into the Shannon Capacity theorem, expressed as $SE = \log_2(1 + SNR)$, which serves as the deterministic training labels for the predictive engine.

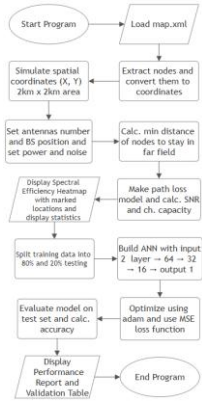


Fig. 3.4 Program's Flowchart

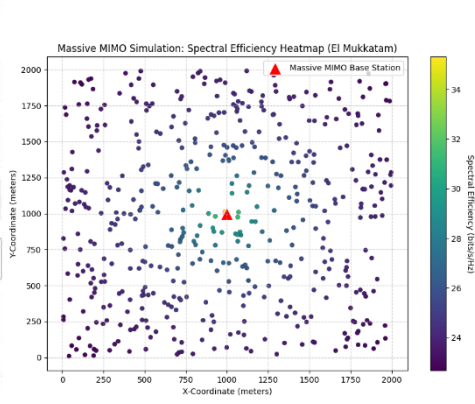


Fig. 3.5 Base Station AI Location Prediction

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--- AI Network Performance Report ---
Location: (1500, 800)
Distance from BS: 538.52 meters
Predicted Spectral Efficiency (SE): 25.60 bps/Hz

Validation Table: Mathematical Calculations vs. AI Predictions (El Mukkatam):
Location (X, Y)  Ground Truth (Math SE)  AI Predicted SE  Accuracy (%)
0  (41, 8)  22.682  22.903000  99.823
1  (1233, 1386)  25.901  26.180000  98.922
2  (142, 1045)  24.050  23.844999  99.149
3  (1278, 1686)  24.474  24.507000  99.863
4  (194, 450)  23.683  23.464001  99.877

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Fig. 3.6 AI Performance Test Using Random Locations & Validation Table

4. MIMO Array Antenna

4.1 Design of MIMO Antenna

The bandwidth of the antenna focuses on 3.3–3.8GHz band; specifically operating frequency of 3.6GHz, to include the Sub-6 GHz 5G operating range for mobile communications. Adding resonant ring patches and cuts in the ground is the main factor that will provide the exact frequency we need and will provide even better return loss values.

We chose the dielectric substrate to be **FR4 epoxy** with relative permittivity of value $\epsilon_r = 4.3$ and chose the standard thickness of it to be $h = 1.6$ mm.

Using the microstrip rules to calculate dimensions of the substrate for each patch:

$$W = \frac{c}{2f_o \sqrt{\frac{(\epsilon_r + 1)}{2}}} = \frac{3 \times 10^8}{2 \times 3.6 \times 10^9 \sqrt{\frac{(4.3 + 1)}{2}}} = 25.5 \text{ mm}$$

The width and length edge limits per one patch are chosen to be $L_p = W_p = 24.75$ mm after several simulations to obtain the required frequency, the width and length are still in the margin of standard calculations. But we will continue using the actual value obtained from the equations as all that is calculated before the simulation.

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + 12 \left(\frac{h}{W_p} \right) \right)^{-\frac{1}{2}}$$

$$\epsilon_{eff} = \frac{4.3 + 1}{2} + \frac{4.3 - 1}{2} \left(1 + 12 \left(\frac{1.6}{25.5} \right) \right)^{-\frac{1}{2}} = 3.89$$

$$L_{eff} = \frac{c}{2f_o \sqrt{\epsilon_{eff}}} = \frac{3 \times 10^8}{2 \times 3.6 \times 10^9 \sqrt{4.085}} = 21.1 \text{ mm}$$

$$\Delta L = 0.412 \times h \times \frac{(\epsilon_{eff} + 0.3) \left(\frac{W_p}{h} + 0.264 \right)}{(\epsilon_{eff} - 0.258) \left(\frac{W_p}{h} + 0.8 \right)}$$

$$\Delta L = 0.412 \times 1.6 \times 10^{-3} \times \frac{(4.085 + 0.3) \left(\frac{38}{1.6} + 0.264 \right)}{(4.085 - 0.258) \left(\frac{38}{1.6} + 0.8 \right)} = 0.736 \text{ mm}$$

$$L_p = L_{eff} - 2 \times \Delta L = 21.1 - 2 \times 0.736 = 19.6 \text{ mm}$$

Length and width of the patch rings are chosen to be $L_{pr} = W_{pr} = 12.25$ mm to satisfy the wanted band, and to be 1 mm thick.

$L_{sub} \times W_{sub} = 150 \times 75$ mm as this is the average mobile phone dimensions.

For the feeding technique, T-shaped feedline was chosen and the feed width W_f was chosen to be $W_f = 3$ mm following the standard value margin and after several evaluations to obtain the desired operating frequency. And the

feed length to make sure it connects the port with the patch was calculated by subtracting the patch ring from the edges and also leaving a gap between the T-shaped feed and the patch ring equal to $x = 0.25 \text{ mm}$ the result of the feed length is $L_f = 6 \text{ mm}$

$$L_f = \frac{W_p - L_p}{2} - x = \frac{24.75 - 12.25}{2} - 0.25 = 6 \text{ mm}$$

The ground cut center be in alignment to the center of the ring and its chosen dimensions are $L_{cut} \times W_{cut} = 13.25 \times 13.25 \text{ mm}$

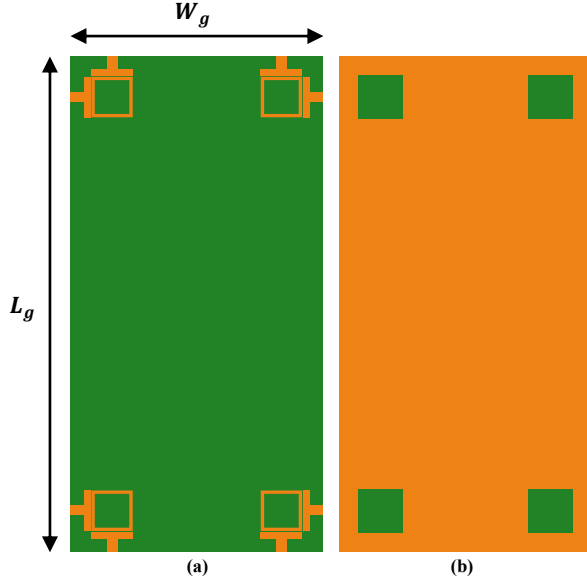


Fig.4.1 Sketch of the MIMO antenna
(a) Front view (b) Back view

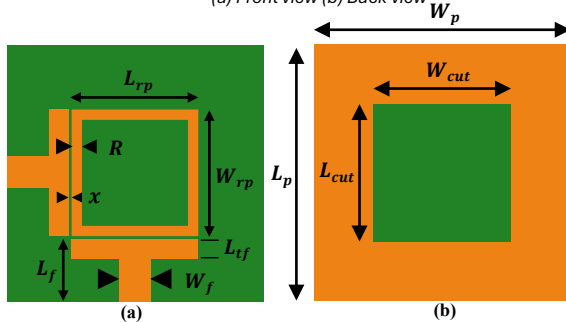


Fig.4.2 Detailed sketch of the MIMO antenna sections
(a) Patch Ring (b) Ground Cut

Parameters symbols	Parameters names	Values (mm)
L_g	Ground length	150
W_g	Ground width	75
L_p	Patch length	24.75
W_p	Patch width	24.75
L_{pr}	Patch ring length	12.25
W_{pr}	Patch ring width	12.25
R	Ring thickness	1
L_f	Feed line length	6
W_f	Feed line width	3
L_{tf}	Feed line T head length	1.9
L_{cut}	Ground cut length	13.25
W_{cut}	Ground cut width	13.25
x	Ring & Feed Gap	0.25
t	Thickness	1.6

Table.4.1 MIMO antenna parameters

4.2 Software Simulation of MIMO Antenna

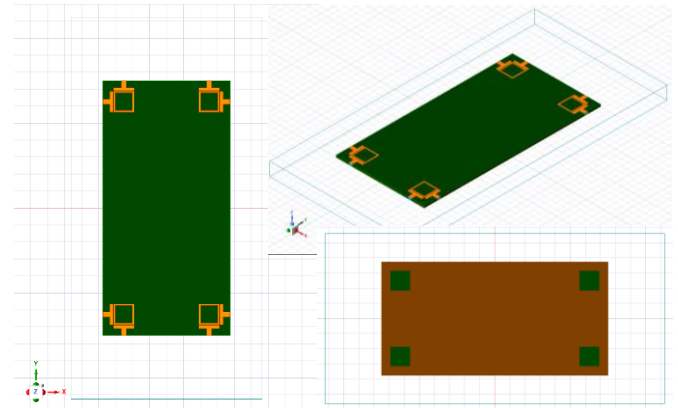


Fig.4.3 MIMO antenna top, orient and bottom view

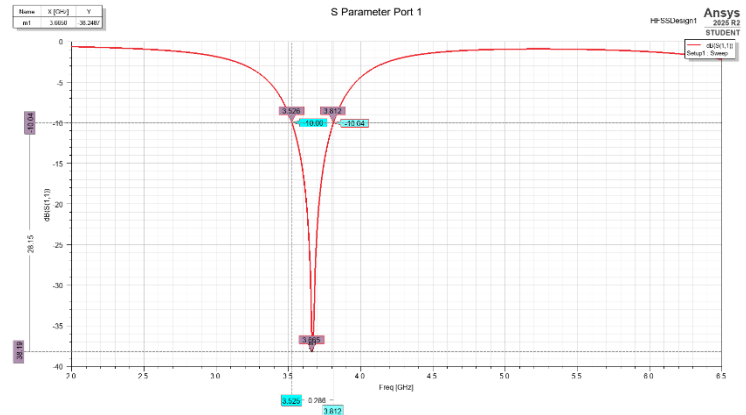


Fig.4.4 1st Port S_{11} Parameter

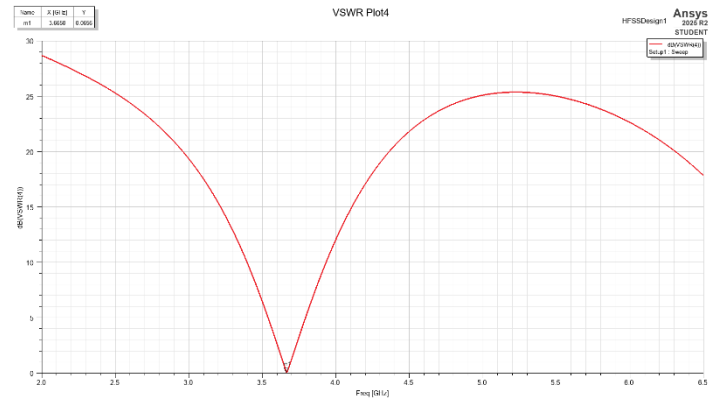


Fig.4.5 4th Port VSWR

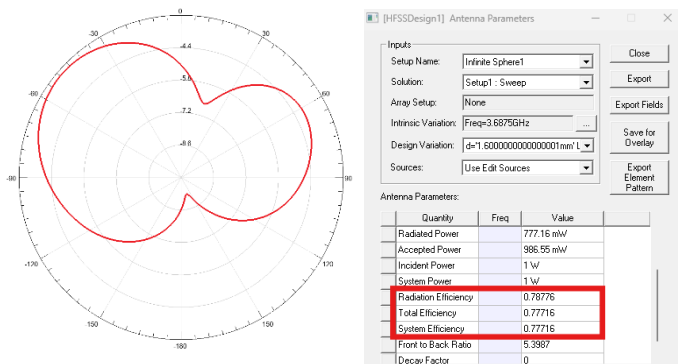


Fig.4.6 MIMO Antenna Radiation Pattern

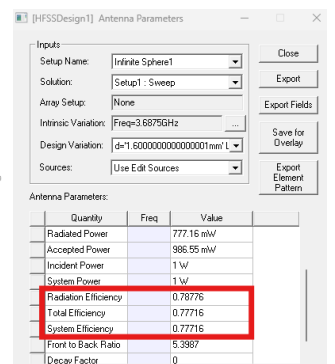


Fig.4.7 MIMO Antenna Efficiency

	Port No.							
	1 st (S1,1)	2 nd (S2,2)	3 rd (S3,3)	4 th (S4,4)	5 th (S5,5)	6 th (S6,6)	7 th (S7,7)	8 th (S8,8)
Frequency (GHz)	3.665	3.665	3.665	3.665	3.665	3.676	3.687	3.665
S- Parameter (dB)	-38.24	-37.87	-39.84	-48.46	-42.21	-41.48	-35.67	-41.97

Table.4.2 S-Parameter VS frequency simulation results comparison for all ports

4.3 MIMO Antenna Hardware Fabrication

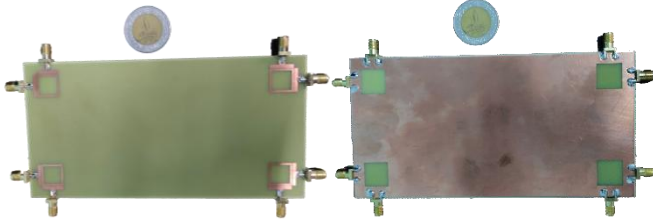


Fig.4.8 Fabricated MIMO Antenna Top and Bottom View

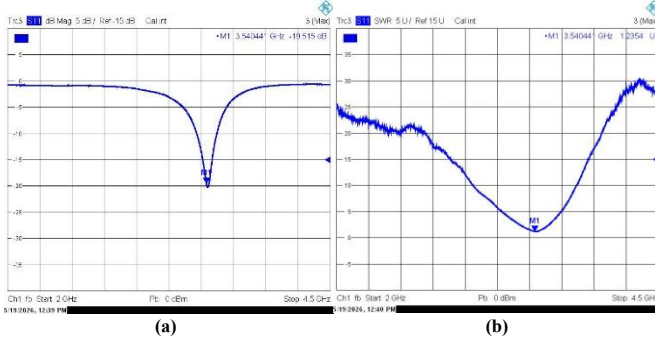


Fig.4.9 Fabricated MIMO Antenna 1st Port Graph Results
(a) S_{11} (b) VSWR

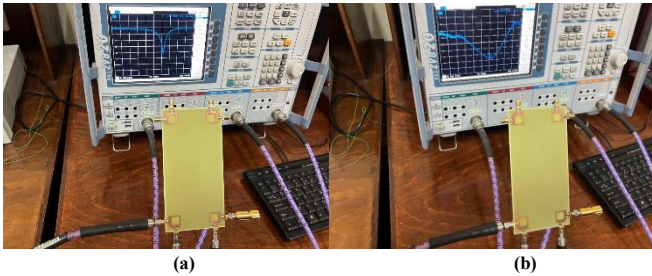


Fig.4.10 Fabricated MIMO Antenna 1st Port results on O-scope
(a) S_{11} (b) VSWR

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